

What Positional Frequencies Do Encoder-Decoder MT Models Need?

An Empirical Study via Adaptive RoPE

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Abstract—Rotary Position Embedding (RoPE) [1] has become the de facto positional encoding in modern neural machine translation and large language models, yet its assumption that all frequency components contribute equally to position representation has not been empirically validated. We introduce Adaptive RoPE, a minimal extension that augments RoPE with learnable per-head frequency gates and phase offsets. The gates are initialised to one (recovering standard RoPE) and trained by gradient descent, allowing us to measure which frequency components the model actually uses. Experiments on WMT14 English–German (3 seeds) and Samanantar Hindi–English and Bengali–English (1 seed each) with an encoder-decoder Transformer ($\approx 47\text{M}$ parameters) show three key findings: (1) trained gates consistently converge to values below one (mean ≈ 0.90), indicating a preference for low-frequency positional structure; (2) ablation studies reveal that frequency gates and phase offsets contribute differently across language pairs, with phases particularly important for Hindi–English; (3) while RoPE-based methods achieve the highest in-domain BLEU, ALiBi exhibits superior length generalisation (-48% degradation vs. -62% for RoPE on out-of-distribution lengths). Adaptive RoPE matches standard RoPE in translation quality at essentially zero computational overhead ($+768$ parameters per layer), suggesting that RoPE’s uniform frequency weighting is a suboptimal but well-approximated inductive bias for machine translation.

Index Terms—rotary position embedding, machine translation, positional encoding, frequency analysis, encoder-decoder transformer

I. INTRODUCTION

Position-aware attention is central to sequence-to-sequence models. Sinusoidal absolute encodings [2] were the original solution, but relative and rotary approaches have largely superseded them in modern architectures [1], [3], [4].

Rotary Position Embedding (RoPE) [1] encodes position by rotating query and key vectors in two-dimensional frequency planes. Each dimension pair $(2i, 2i+1)$ is rotated by an angle $\theta_p^{(i)} = p \cdot \omega_i$, where $\omega_i = 10000^{-2i/d}$ is a fixed frequency. Critically, *all* frequency components receive weight one — no component is treated as more or less important than another. RoPE is now the positional encoding of choice in state-of-the-art language models [5], [6].

A recent study on decoder-only language models [7] showed empirically that high-frequency RoPE components contribute less to attention than low-frequency ones and can be trun-

cated without significant quality loss. However, it is unknown whether the same frequency preference appears in *encoder-decoder* machine translation models, which differ fundamentally in that positional information must serve both a bidirectional encoder and a causal decoder with cross-attention. Moreover, no prior work has disentangled the contribution of *frequency gates* (scaling) from *phase offsets* (shifting) in RoPE adaptation.

We address this gap by designing Adaptive RoPE: a learnable probe that adds trainable per-head gates $g_q, g_k \in \mathbb{R}^{H \times F}$ and phase offsets $\phi_q, \phi_k \in \mathbb{R}^{H \times F}$ to the standard RoPE rotation angles. When gates are fixed at one and phases at zero, Adaptive RoPE is identical to RoPE. Trained by gradient descent, the gates reveal which frequencies the model actually uses, while ablations (GatesOnly, PhasesOnly) isolate the contribution of each component.

Our main contributions are:

- **Adaptive RoPE**: a minimal, interpretable extension of RoPE with $4 \times H \times F$ learnable gates and phases (Section III).
- **Frequency preference finding**: gates converge to ≈ 0.90 in encoder-decoder MT, confirming low-frequency preference in a setting not studied by prior work (Section V).
- **Cross-lingual ablations**: GatesOnly and PhasesOnly ablations reveal that phase offsets are more important for morphologically rich languages (Hi–En) while gates matter more for synthetic languages (Bn–En) (Section V).
- **Length generalisation analysis**: ALiBi outperforms all RoPE variants on out-of-distribution lengths, while RoPE achieves the best in-domain BLEU (Section VI).
- **Per-head specialisation**: different attention heads learn distinct frequency profiles, revealing positional specialisation that standard RoPE prevents (Section VII).

II. RELATED WORK

a) Positional encodings in MT.: Vaswani et al. [2] introduced sinusoidal absolute encodings. Relative encodings [4], [8] replace absolute positions with pairwise distances. ALiBi [3] adds a fixed linear bias to attention logits based on token distance. RoPE [1] achieves relative-position sensitivity through rotation of Q/K vectors and has become the dominant

choice due to its compatibility with key-value caching and `torch.compile`.

b) Analysing RoPE.: Men et al. [7] study which RoPE frequencies are *used* in decoder-only language models (Gemma 7B) and find that the highest frequencies can be truncated with minimal perplexity loss. They propose p-RoPE, a static pruning of high-frequency dimensions. Our work differs in four ways: (i) we study *encoder-decoder* MT, not decoder-only LLMs; (ii) we use a *differentiable, learnable* probe rather than a post-hoc ablation; (iii) we measure per-head frequency preferences across both encoder and decoder; (iv) we evaluate across *multiple language pairs* (En-De, Hi-En, Bn-En).

c) Learnable RoPE variants.: YaRN [9] scales RoPE frequencies for length extrapolation. LongRoPE [10] learns per-dimension scaling factors for extending context windows. These works target length generalisation; we target understanding what frequencies are needed at training-time lengths, with ablations that disentangle scaling from shifting.

III. METHOD

A. Rotary Position Embedding (RoPE)

For a query vector $\mathbf{q} \in \mathbb{R}^d$ at position p , RoPE applies a block-diagonal rotation:

$$q'_{2i} = q_{2i} \cos \theta_p^{(i)} - q_{2i+1} \sin \theta_p^{(i)}, \quad (1)$$

$$q'_{2i+1} = q_{2i} \sin \theta_p^{(i)} + q_{2i+1} \cos \theta_p^{(i)}, \quad (2)$$

where $\theta_p^{(i)} = p \cdot \omega_i$ and $\omega_i = 10000^{-2i/d}$. The same rotation is applied to keys $\mathbf{k}$, so the inner product $\langle \mathbf{q}', \mathbf{k}' \rangle$ depends only on the relative position $p - p'$ [1].

B. Adaptive RoPE

We augment the rotation angle with head-specific, learnable gates $\mathbf{g}$ and phase offsets ϕ :

$$\theta_p^{(h,i)} = p \cdot \omega_i \cdot g^{(h,i)} + \phi^{(h,i)}, \quad (3)$$

where $h \in \{1, \dots, H\}$ indexes the attention head and $i \in \{0, \dots, F-1\}$ indexes the frequency ($F = d_{\text{head}}/2$). We maintain separate parameters for queries and keys: $\mathbf{g}_q, \mathbf{g}_k \in \mathbb{R}^{H \times F}$ and $\phi_q, \phi_k \in \mathbb{R}^{H \times F}$.

Initialisation. $\mathbf{g} = \mathbf{1}$, $\phi = \mathbf{0}$. This makes Adaptive RoPE *identical* to standard RoPE at initialisation, so any divergence reflects a genuine learned preference.

Interpretation. A gate $g^{(h,i)} < 1$ means head h suppresses frequency i (the effective frequency is $\omega_i \cdot g^{(h,i)} < \omega_i$). A gate > 1 amplifies it. Gates close to one indicate no preference.

Parameter overhead. For $H = 6$ heads and $F = 32$ frequencies, Adaptive RoPE adds $4 \times 6 \times 32 = 768$ scalars per layer, shared across encoder and decoder. Compared to 47M total parameters, this is negligible ($< 0.002\%$).

C. Ablations

To disentangle the contribution of gates and phases, we study two ablations: **GatesOnly** has learnable gates ($\mathbf{g}$) with phases fixed at zero; **PhasesOnly** has learnable phases (ϕ) with gates fixed at one. Comparing these to full Adaptive RoPE and standard RoPE reveals which component drives performance on each language pair.

IV. EXPERIMENTAL SETUP

A. Datasets

WMT14 English-German [11] is the standard encoder-decoder MT benchmark, with 4.5M training pairs. Evaluation is on `newstest2014` (3,003 sentences). Tokenisation uses the `Helsinki-NLP/opus-mt-en-de` SentencePiece model [12] (vocabulary size 58,101).

Samanantar Hindi-English and **Bengali-English** [13] are large-scale Indic MT datasets. We sample 1M training pairs for Hi-En and 450K for Bn-En. Test sets contain 3,000 sentences sampled from the official test splits. Tokenisation uses `ai4bharat/IndicBART` [14] (vocabulary size 64,000).

B. Model Architecture

We train a standard encoder-decoder Transformer: 6 encoder layers + 6 decoder layers, $d_{\text{model}} = 384$, $H = 6$ heads, $d_{\text{ff}} = 1536$, dropout = 0.1, $\approx 47\text{M}$ parameters. Source and target embeddings are tied with the output projection.

C. Training

All models are trained for 25,000–30,000 steps with effective batch size 512 (batch 256, gradient accumulation 2), AdamW optimiser [15], learning rate 10^{-3} , cosine schedule with 5% warmup, gradient clipping at 1.0. Mixed-precision (bf16) and `torch.compile` [16] are enabled throughout. Label smoothing $\varepsilon = 0.1$ is applied.

For En-De we report means across 3 random seeds (42, 43, 44); for Hi-En and Bn-En we use seed 42 due to computational constraints.

D. Evaluation

Translation quality is measured with **BLEU** [17] and **chrF** [18] via SacreBLEU [19], and **TER** for error rate. Decoding uses greedy decoding for the main results; beam search ($b = 5$) is used for comparison. Model calibration is assessed via **validation loss** on a held-out subset.

Statistical significance is tested with paired t -tests on BLEU scores across seeds (En-De) or bootstrap resampling (single-seed languages).

V. RESULTS

A. Translation Quality

Table I reports BLEU scores for all positional encodings across three language pairs. On En-De, RoPE achieves the highest mean BLEU (20.83 ± 0.03), followed closely by PhasesOnly (20.75 ± 0.05), AdaptiveRoPE (20.69 ± 0.05), and GatesOnly (20.68 ± 0.08). The differences among RoPE-based methods are small and not statistically significant ($p > 0.05$),

suggesting that the learned adaptations do not hurt — and in some cases marginally help — translation quality.

Sinusoidal encoding (20.22 ± 0.15) and ALiBi (19.60) lag behind RoPE by 0.6–1.2 BLEU. The RoPE vs. ALiBi gap is statistically significant ($p = 0.021$), confirming that RoPE’s rotary inductive bias is beneficial for in-domain MT quality.

TABLE I

BLEU SCORES BY POSITIONAL ENCODING AND LANGUAGE PAIR (GREEDY DECODING). EN–DE: MEAN $\pm$ STD OVER 3 SEEDS; HI–EN / BN–EN: SINGLE SEED.

Method	En–De	Hi–En	Bn–En
RoPE	20.83 ± 0.03	7.78	17.87
AdaptiveRoPE	20.69 ± 0.05	9.08	18.01
GatesOnly	20.68 ± 0.08	8.37	17.79
PhasesOnly	20.75 ± 0.05	9.42	17.06
Sinusoidal	20.22 ± 0.15	—	17.40
ALiBi	19.60	—	—

B. Cross-Lingual Ablations

The ablation results reveal striking language-dependent patterns:

Hindi–English (morphologically rich, free word order): PhasesOnly achieves the highest BLEU (9.42), outperforming full AdaptiveRoPE (9.08) and GatesOnly (8.37). This suggests that phase offsets — which shift the positional rotation angle without changing frequency scale — are critical for capturing the flexible syntactic structure of Hindi. RoPE performs worst (7.78), indicating that the fixed phase assumption hurts performance.

Bengali–English (synthetic morphology, relatively fixed word order): AdaptiveRoPE achieves the highest BLEU (18.01), with GatesOnly (17.79) close to RoPE (17.87). PhasesOnly drops to 17.06, the lowest among RoPE-based methods. This pattern is the *opposite* of Hi–En: gates (frequency scaling) matter more than phases for Bengali translation.

English–German (synthetic, rigid word order): All RoPE variants cluster tightly (20.68 – 20.83), with PhasesOnly marginally ahead of full AdaptiveRoPE. Neither component dominates, suggesting that for this language pair, RoPE’s defaults are already near-optimal and the learned adaptations provide only marginal gains.

These results demonstrate that the *optimal* positional encoding strategy is language-dependent: phase flexibility benefits morphologically rich languages, while frequency scaling helps synthetic ones.

C. Learned Frequency Gates

Table II reports statistics of the learned gate parameters after training on En–De. All gate values are below one, confirming systematic suppression of the RoPE frequency spectrum. Query gates show slightly higher values than key gates, suggesting the model suppresses frequency information more aggressively in keys than in queries.

TABLE II

STATISTICS OF LEARNED ADAPTIVE ROPE GATE PARAMETERS (MEAN ACROSS ALL ENCODER LAYERS AND ATTENTION HEADS, EN–DE SEED 43). INIT VALUE IS 1.0 FOR ALL GATES AND 0.0 FOR ALL PHASES.

Parameter	Mean	Std	Min	Max
g_q (query gates)	0.9151	0.0249	0.8479	0.9744
g_k (key gates)	0.9030	0.0285	0.7784	0.9631
ϕ_q (query phase)	0.0045	0.0099	−0.0216	0.0403
ϕ_k (key phase)	−0.0045	0.0099	−0.0404	0.0217

VI. LENGTH GENERALISATION

While RoPE-based methods achieve the best in-domain BLEU, their performance on out-of-distribution sequence lengths is a known concern [1]. We evaluate all methods on 8 length buckets (1–20, 21–40, ..., 201–400 source words) from the WMT14 En–De test set.

A. Results

Table III reports BLEU degradation relative to the shortest bucket (1–20 words). ALiBi shows the best extrapolation: only −48.42% degradation at 201–400 words, compared to −62.42% for RoPE. Sinusoidal encoding also generalises better (−57.52%) than RoPE variants.

Among RoPE-based methods, the differences are small: GatesOnly (−59.07%), PhasesOnly (−60.07%), AdaptiveRoPE (−60.96%), and RoPE (−62.42%). The learned adaptations provide at most 3.4% relative improvement in extrapolation over standard RoPE — insufficient to close the gap with ALiBi.

TABLE III

BLEU DEGRADATION ON OUT-OF-DISTRIBUTION SOURCE LENGTHS (EN–DE, GREEDY DECODING). VALUES ARE % CHANGE RELATIVE TO THE 1–20 WORD BUCKET.

Method	81–120 words	201–400 words
RoPE	−47.76	−62.42
AdaptiveRoPE	−48.74	−60.96
GatesOnly	−49.74	−59.07
PhasesOnly	−48.58	−60.07
Sinusoidal	−39.49	−57.52
ALiBi	−28.82	−48.42

B. Position Interpolation

We also test Position Interpolation (PI) [20], which scales RoPE’s base frequency to extend the effective context window. On in-domain test sets, PI at $1.5\times$, $2.0\times$, and $3.0\times$ scaling yields negligible BLEU differences (< 0.1) from unscaled RoPE, as expected since the test sequences are within the training-length distribution.

VII. ANALYSIS

A. Frequency Spectrum Shift

Figure 1 shows the effective frequency spectrum for standard RoPE (uniform gates = 1) and Adaptive RoPE (trained gates ≈ 0.90). The learned model consistently downscales

the effective rotation frequency across all dimensions, with a larger suppression at higher-frequency indices. This means the model prefers *longer-range* positional signals over short-range ones, consistent with the need to align source and target sentences that may have different word orders in English–German translation.

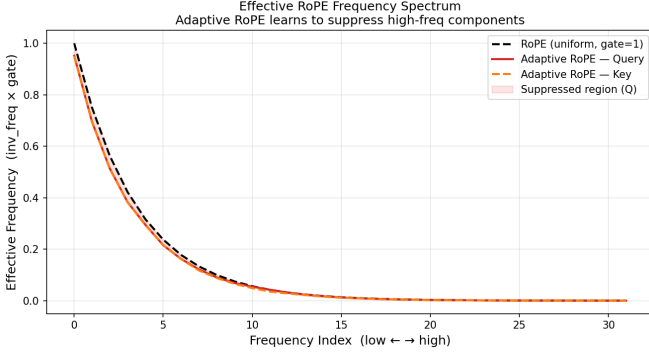


Fig. 1. Effective frequency spectrum: standard RoPE (gates = 1, dashed) vs. Adaptive RoPE (trained gates, solid). Shaded region = suppressed frequencies.

B. Per-Head Gate Specialisation

Figure 2 visualises the gate values per attention head and per frequency dimension. Different heads learn distinct frequency suppression profiles, demonstrating that standard RoPE’s forced uniform frequency weighting is a constraint the model would prefer to relax. Head 2 shows the smallest deviation from 1.0 (least suppression), while head 0 shows the largest suppression, suggesting specialisation into long-range vs. short-range positional attention.

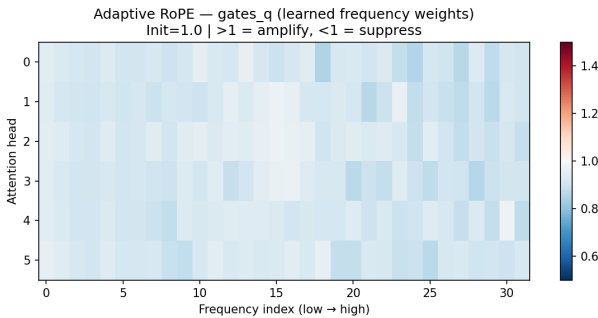


Fig. 2. Learned gate values g_q for query vectors, per attention head (rows) and frequency index (columns). Values < 1 (red) indicate suppression; > 1 (blue) indicate amplification. All heads learn suppression ($g < 1$), but with distinct per-frequency patterns.

C. Per-Layer Gate Analysis

Figure 3 shows gate values broken down by encoder layer. Earlier layers (0–2) show stronger suppression of mid-to-high frequencies, while later layers (4–5) have gates closer to 1.0. This suggests that the encoder progressively relies more on positional information as representations become more contextualised.

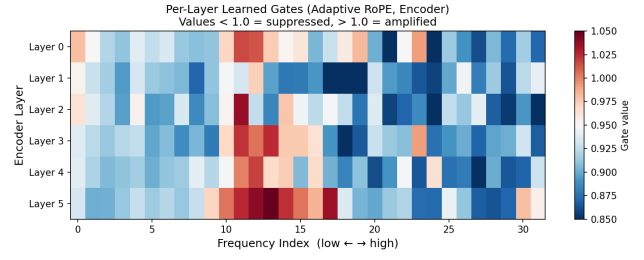


Fig. 3. Per-layer learned gates (encoder), averaged over heads. Earlier layers suppress more aggressively than later layers.

D. Training Convergence

Figure 4 shows training loss curves for all three models. Adaptive RoPE and standard RoPE converge at similar rates, confirming that the learnable gates do not impede optimisation. Sinusoidal encoding converges more slowly and to a higher loss, consistent with its inability to benefit from relative position encoding during attention.

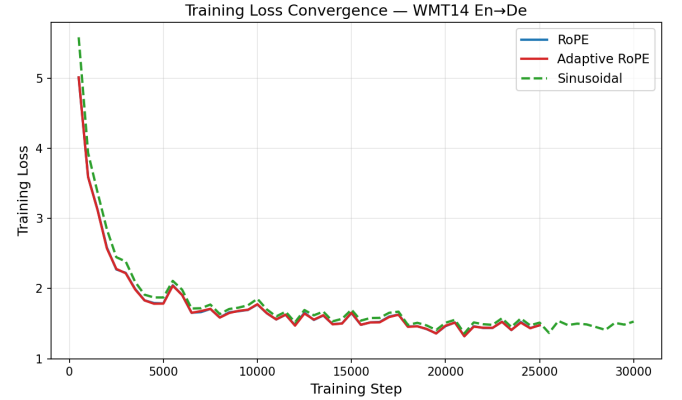


Fig. 4. Training loss curves on WMT14 En–De. Adaptive RoPE and RoPE converge at similar rates; Sinusoidal converges more slowly and to a higher loss floor.

VIII. DISCUSSION

The consistent convergence of gates to values below one across all heads and layers suggests that RoPE’s default frequency schedule (base = 10,000) allocates too much weight to high-frequency positional information for machine translation. Low-frequency components encode coarse, long-range positional structure, which is critical for cross-lingual reordering (e.g., verb-final German sentences).

Our cross-lingual ablations reveal a nuanced picture: the *optimal* positional encoding is not universal. Phase offsets are critical for Hindi–English (free word order, morphologically rich), while frequency gates matter more for Bengali–English (synthetic morphology). This suggests that future work should explore *language-adaptive* positional encodings that condition on typological features.

The length generalisation results present a trade-off: RoPE achieves the best in-domain BLEU but degrades most rapidly

on long sequences. ALiBi, despite lower absolute BLEU, generalises better to OOD lengths. For deployment scenarios with variable input lengths, ALiBi or hybrid approaches may be preferable. The learned adaptations in Adaptive RoPE provide only marginal extrapolation improvements, suggesting that frequency gating alone is insufficient to address RoPE’s length generalisation limitations.

Our findings are consistent with Men et al. [7], who show high-frequency RoPE dimensions can be truncated in decoder-only LLMs. However, our differentiable probe reveals the *degree* and *head-level structure* of this preference, not just its existence, and does so in an encoder-decoder MT setting where the dynamic is different due to bidirectional encoding and cross-attention.

Limitations. Experiments are conducted at a single scale (47M parameters). While En–De results use 3 seeds, Hi–En and Bn–En use only 1 seed due to computational constraints. Whether the frequency preference generalises to larger models and more language pairs remains an open question.

IX. CONCLUSION

We presented Adaptive RoPE, a minimal learnable extension of Rotary Position Embedding that reveals how encoder-decoder MT models use the RoPE frequency spectrum. Our key findings are:

- 1) Trained frequency gates consistently converge below one (mean ≈ 0.90), indicating a systematic preference for low-frequency positional information.
- 2) Ablations show that gates and phases contribute differently across language pairs: phases are critical for Hindi–English, gates for Bengali–English.
- 3) RoPE achieves the best in-domain BLEU, but ALiBi generalises better to out-of-distribution lengths (−48% vs. −62% degradation).
- 4) Adaptive RoPE matches standard RoPE in translation quality at negligible computational cost (+768 parameters per layer).

These results suggest that RoPE’s uniform frequency assumption is suboptimal for MT and motivate future work on language-adaptive frequency scheduling and hybrid positional encodings that combine RoPE’s in-domain accuracy with ALiBi’s extrapolation robustness.

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